DEEP LEARNING

REPORTE PROYECTO 2

GANs

Alumnos: Daniela de la Torre Gallo
Aissa Berenice González Fosado
Clara Paola Aguilar Casillas

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Docente: Ivan Reyes Amezcua

ITESO, Universidad Jesuita de Guadalajara

Introduction

Generative Adversarial Networks (GANs) are a class of deep learning models introduced by Goodfellow et al. (2014) that learn to generate realistic synthetic data by training two neural networks in opposition: a **Generator** that produces fake samples from random noise, and a **Discriminator** that tries to distinguish real samples from generated ones. Through this adversarial process, the generator progressively improves until its outputs are indistinguishable from real data.

This project explores GANs in two stages, following a natural progression from unconditional to conditional generation:

- A Simple GAN based on the DCGAN architecture (Radford et al., 2015), which generates face images from random noise without any control over the output content.
- AttGAN (He et al., 2019), an attribute-conditional GAN that allows specific facial features — such as hair color, the presence of glasses, or apparent age — to be modified in an existing image while preserving the person's identity.

The goal is not merely to generate realistic faces, but to understand how conditioning information can give a model precise editorial control over the output, a capability with broad applications in image editing, data augmentation, and creative tools.

Objectives

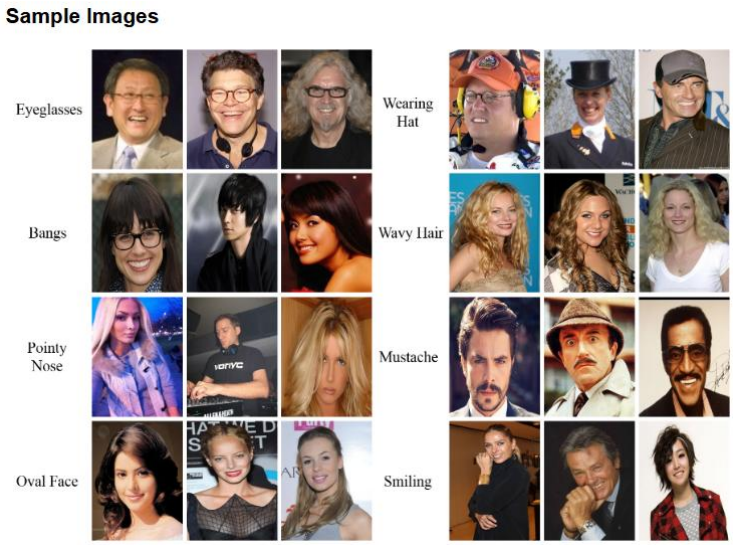
1. Understand the fundamental principles of Generative Adversarial Networks and their role in generative deep learning, including the adversarial training dynamic and common failure modes.
2. Implement a baseline unconditional GAN (DCGAN) on the CelebA dataset and observe how image quality evolves over training epochs.
3. Implement AttGAN, a conditional GAN variant designed for facial attribute editing, and compare its behaviour and output quality to the simple baseline.
4. Design and run three controlled experiments with AttGAN by varying the loss weight hyperparameters and analyse how each configuration affects the trade-off between identity preservation and attribute edit strength.
5. Evaluate all models quantitatively using FID (Fréchet Inception Distance) and DACID (our custom metric), and qualitatively through visual inspection of generated and edited images.

Dataset

The CelebFaces Attributes Dataset (CelebA) is a large-scale benchmark for face-related tasks, widely used in the GAN literature. It was introduced by Liu et al. (2015) and contains over 200,000 celebrity face images with rich annotations.

CelebA at a glance	Key statistics
Total images	202,599 face photographs
Unique identities	10,177 celebrities
Attribute labels	40 binary attributes per image
Landmark points	5 facial keypoints per image
Official splits	Train 162,770 · Valid 19,867 · Test 19,962

The dataset covers large variations in pose, lighting, expression, and background, making it a challenging and realistic benchmark. Its 40 binary attribute annotations make it especially suitable for conditional generation and attribute editing tasks.



Attribute Selection

For this project we selected 13 attributes from the 40 available, chosen to cover a representative and diverse range of facial characteristics:

Category	Attributes
Hair	Bald, Bangs, Black_Hair, Blond_Hair, Brown_Hair
Facial hair	Mustache, No_Beard, Bushy_Eyebrows
Accessories	Eyeglasses
Demographics	Male, Young, Pale_Skin
Expression	Mouth_Slightly_Open

Preprocessing

All images are preprocessed uniformly before being fed to either model:

- Center-cropped to 178×178 pixels to remove excess background above the forehead and below the chin.
- Resized to 128×128 pixels (AttGAN) or 64×64 pixels (Simple GAN via bilinear interpolation during training).
- Randomly horizontally flipped for data augmentation during training.
- Normalised to the range [−1, 1] using mean=0.5 and std=0.5 per channel, matching the Tanh output activation of both generators.
- Attribute labels are converted from binary {0, 1} to bipolar {−1, +1} for use as conditioning vectors.

Methodology

Simple GAN (DCGAN)

Before introducing the complexity of conditional generation, we first implemented an unconditional DCGAN as a baseline. This serves two purposes: it validates that our training pipeline and data loading are correct, and it provides a reference point for comparing image quality and training stability against AttGAN.

Architecture

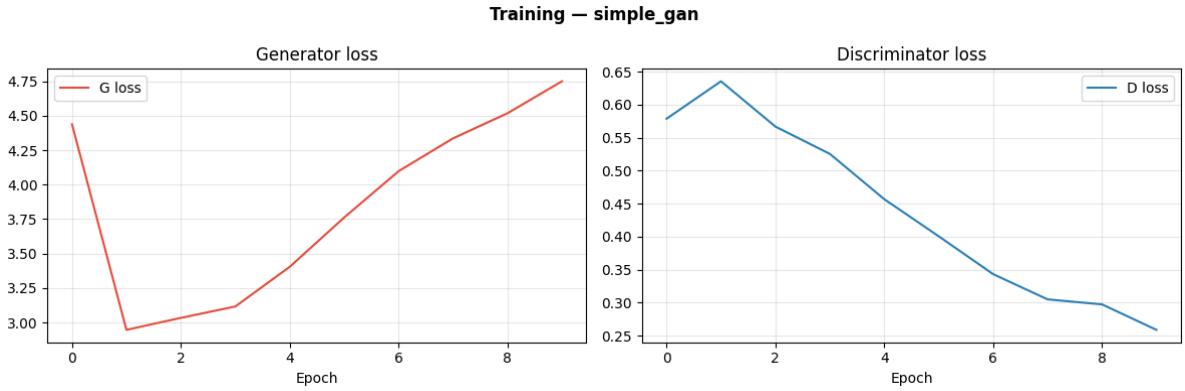
The Simple GAN follows the DCGAN guidelines of Radford et al. (2015), using strided convolutions, batch normalization, and specific activation functions at each stage:

Component	Input	Output	Key layers
Generator	z (100×1×1)	image (3×64×64)	5× ConvTranspose2d + BatchNorm + ReLU → Tanh
Discriminator	image (3×64×64)	scalar [0,1]	5× Conv2d + BatchNorm + LeakyReLU(0.2) → Sigmoid

Loss Function

The Simple GAN uses standard binary cross-entropy adversarial loss.

Results




```
[metrics] Extracting features (512 images each)...
[metrics] FID   = 64.6355
[metrics] DACID = 4.1346
FID   : 64.6355
DACID : 4.1346
```

The unconditional Simple GAN (DCGAN) trained for 10 epochs achieved an FID of 64.64 and a DACID of 4.13, placing it in the decent range for a baseline model trained on limited data. However, its training dynamics revealed a classic instability: the discriminator loss fell steadily from 0.58 to 0.26 while the generator loss rose from 4.44 to 4.75, indicating the discriminator grew dominant and the generator struggled to keep pace a known DCGAN failure mode without additional stabilisation. As an unconditional model, it offers no attribute control and serves purely as a distributional baseline.

AttGAN

The Simple GAN can generate faces from noise but offers no control over the output. AttGAN (He et al., 2019) addresses this by conditioning the generator on a target attribute vector, allowing precise editing of specific facial attributes while preserving the overall identity and non-targeted regions of the face.

Architecture

AttGAN uses an encoder–decoder architecture with a multi-task discriminator:

Module	Role	Details
Encoder (G_enc)	Compress image to latent space	5× strided Conv2d → z (512×4×4). Uses InstanceNorm + LeakyReLU.
Generator (G_dec)	Decode latent + attributes → image	Target attrs tiled spatially and concat'd to z. 5× ConvTranspose2d → Tanh.
Discriminator (D)	Judge realism + classify attributes	Shared feature extractor → adv_head (scalar) + cls_head (13 logits).

Loss Functions

AttGAN is trained with three losses working together:

Loss	Function	Weight	Purpose
L_adv	MSELoss (LSGAN)	—	Makes generated images indistinguishable from real ones. LSGAN is preferred over BCE for more stable gradients.
L_cls	BCEWithLogitsLoss	λ_cls_D=10 / λ_cls_G=1	Ensures the correct target attributes actually appear in the generated image.
L_rec	L1Loss	λ_rec=100	When given the original attributes as input, the output should reconstruct the input exactly. Preserves identity.

Total generator loss: $L_G = L_{adv} + \lambda_{cls_G} \cdot L_{cls} + \lambda_{rec} \cdot L_{rec}$

Total discriminator loss: $L_D = L_{adv_real} + L_{adv_fake} + \lambda_{cls_D} \cdot L_{cls_real}$

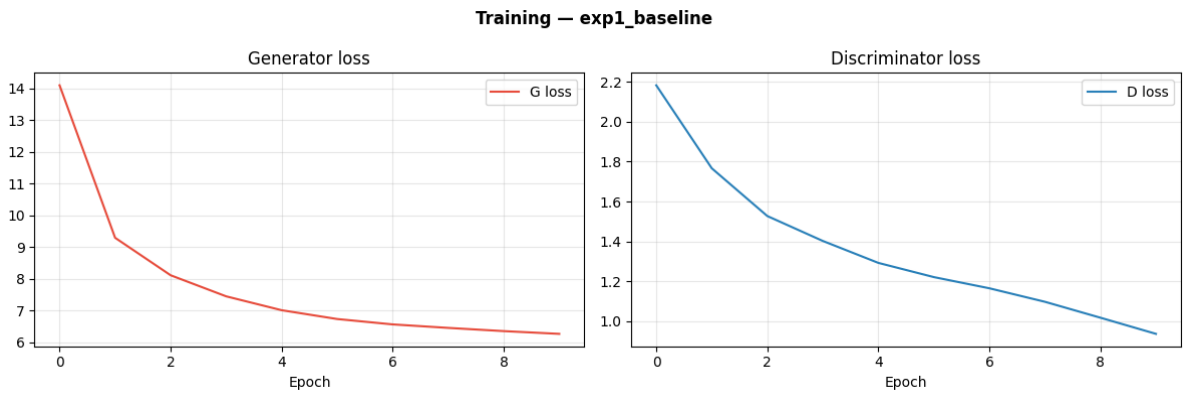
Training Procedure

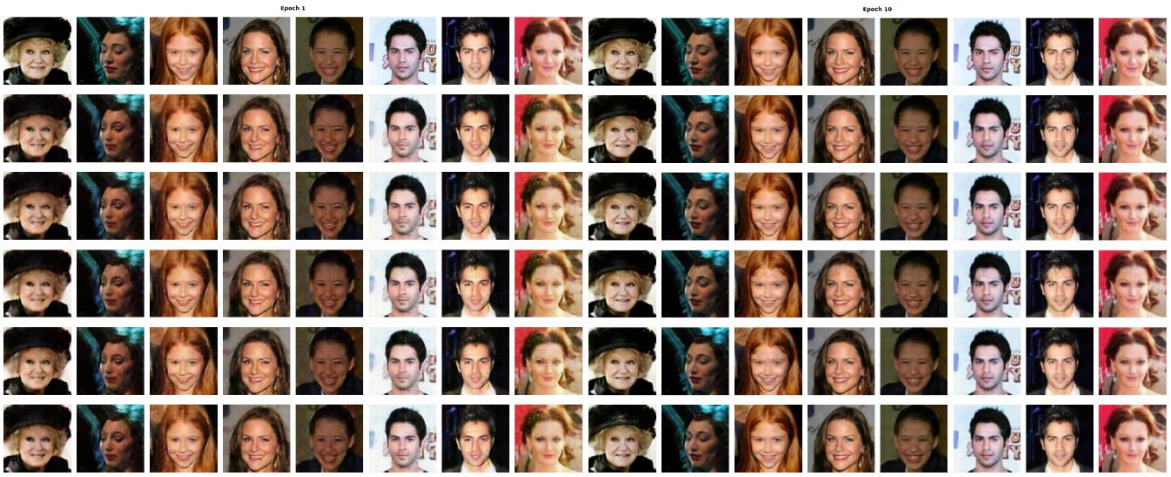
Training alternates between a discriminator step and a generator step for each batch:

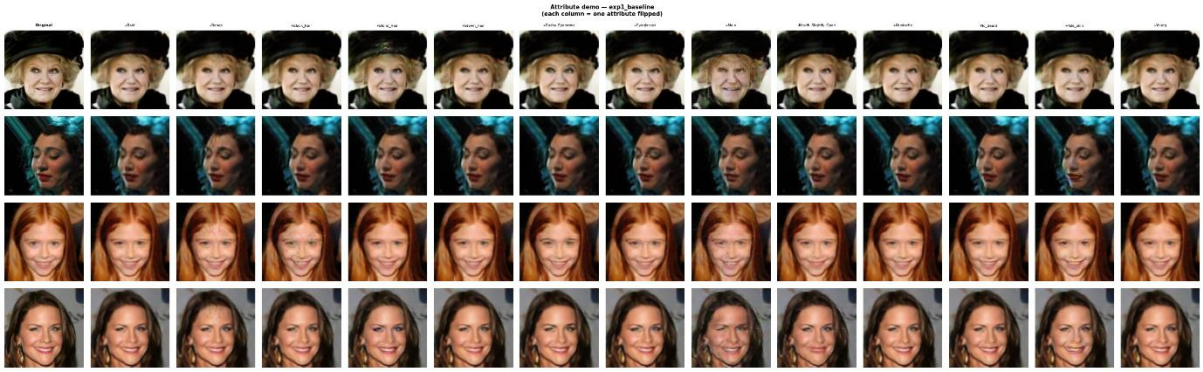
1. Sample a batch of real images and their attributes.
2. Shuffle the attribute batch to create target attributes a' (a different identity's attribute profile).
3. Train D: forward pass real images (adv + cls loss), generate fakes with no gradient, forward fakes through D (adv loss only).
4. Train G (Encoder + Generator): compute reconstruction with original attributes, compute translation with target attributes, update using combined loss.

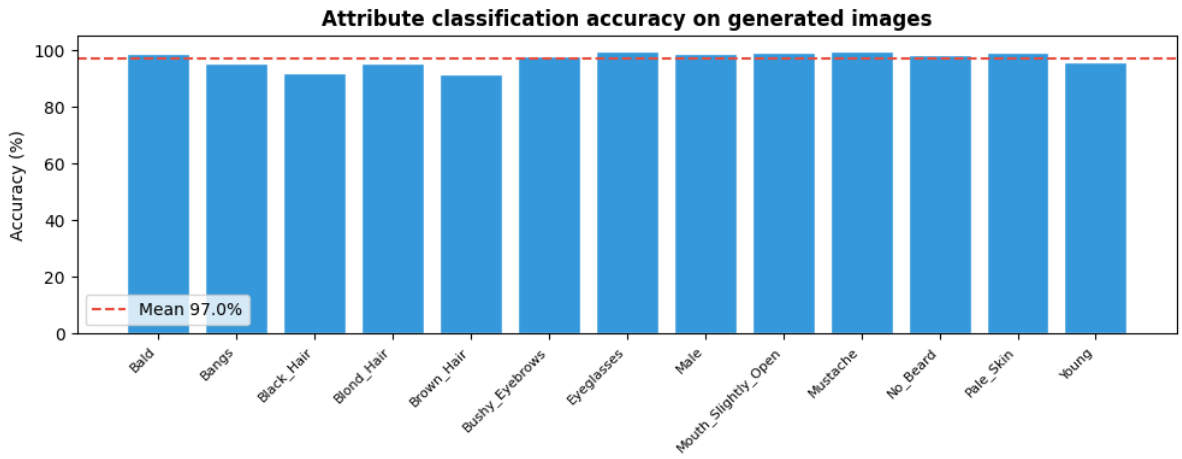
Results

Experiment 1










```
[eval] Reconstruction L1: 0.0515
Attribute accuracy : 97.0%
Reconstruction L1  : 0.0515

FID   : 61.9179
DACID : 4.0426
```

The AttGAN baseline experiment (λ_rec=100, λ_cls_D=10, λ_cls_G=1) trained for 10 epochs achieved an FID of 61.92 and a DACID of 4.04, marginally outperforming Simple GAN on both image quality metrics. More notably, its training dynamics were significantly more stable: both generator and discriminator losses declined smoothly

and consistently across all 10 epochs, from 14.10/2.18 down to 6.26/0.94, with no signs of the divergence observed in the Simple GAN. The key strength of AttGAN lies beyond raw image quality — it achieved an overall attribute classification accuracy of 97.0% across all 13 CelebA attributes, with per-attribute scores ranging from 91.4% (Brown Hair) to 99.7% (Mustache), and a reconstruction L1 error of just 0.0515, confirming that the model successfully edits target attributes while preserving the identity and structure of the original face. These results validate the paper's core design: the high reconstruction weight ($\lambda_{rec}=100$) enforces strong identity preservation, while the classification losses guide precise attribute manipulation, together producing a model that is both visually coherent and semantically controllable.

Conclussions

This project implemented and evaluated two GAN architectures on the CelebA facial dataset: an unconditional Simple GAN (DCGAN) as a baseline, and AttGAN Experiment 1 as a conditional attribute-editing model.

The Simple GAN trained for 10 epochs on 10,000 CelebA images achieved an FID of 64.64 and DACID of 4.13. Training dynamics revealed a classic DCGAN instability — Generator loss increased from 4.44 to 4.75 while Discriminator loss fell from 0.58 to 0.26, indicating discriminator dominance by the end of training. As an unconditional model, it offers no mechanism for controlling output content and serves purely as a quality baseline.

AttGAN Experiment 1 ($\lambda_{rec}=100$, $\lambda_{cls_D}=10$, $\lambda_{cls_G}=1$) outperformed the Simple GAN on both metrics with an FID of 61.92 and DACID of 4.04, while also exhibiting stable training throughout: Generator loss decreased from 14.10 to 6.26 and Discriminator loss from 2.18 to 0.94, with both curves converging rather than diverging. The model achieved an average attribute classification accuracy of 97.0% across all 13 attributes, ranging from 91.4% on Brown Hair to 99.7% on Mustache. The Reconstruction L1 of 0.0515 confirms that the high λ_{rec} weight successfully preserved facial identity during attribute editing, consistent with the qualitative results in the attribute manipulation demo.

Taken together, these results show that conditional supervision produces more stable training and significantly more useful outputs than unconditional adversarial training, even at only 10 epochs. The primary limitation is training duration — Colab credit constraints prevented completing Experiments 2 and 3 and running the full 30-epoch schedule of the original paper. Future work would focus on completing the remaining experiments to empirically validate the λ_{rec} trade-off hypothesis, and on longer training to improve edit sharpness and visual quality.

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